



Convolutional neural network based capacity estimation using random segments of the charging curves for lithium-ion batteries

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ABSTRACT

Capacity estimation is an essential task for battery management systems to ensure the safety and reliability of lithium-ion batteries. Considering the uncertainty of charging and discharging behavior in practical usage, this paper presents a one-dimensional convolutional neural network (1D CNN)-based method that takes random segments of charging curves as inputs to perform capacity estimation for lithium-ion batteries. To improve the robustness and accuracy of the proposed 1D CNN network, a linear decreasing weighted particle swarm optimization algorithm is utilized to optimize the partial hyperparameters of neural network. Experimental data from two sets of batteries with different nominal capacities are employed for verification purpose. It is proved that the proposed method is feasible to provide accurate estimations on capacity degradation for both kinds of batteries. Furthermore, effects of length and relative position of segments on the capacity estimation are also investigated. The analysis results show that a more precise estimation of the battery capacity is prone to be obtained from the segment with a longer length or lower initial SOC.

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1. Introduction

Compared to conventional batteries, lithium-ion batteries are well-known by their high energy densities, low costs and long cycle life [1–3]. Such advantages contribute to the successful use of lithium-ion batteries in many fields including electric vehicles (EV), hybrid electric vehicles (HEV), etc., but on the other hand result in serious reliability issues that may cause disastrous consequences. To address these problems, it is very important to develop effective methods to detect and monitor lithium-ion batteries' degradation throughout their service life [4]. As an important indicator of the state of health, capacity is often used to describe the electric charge stored in battery that can be released in fully discharged cycle, and therefore gains much attention in reliability analysis of the batteries [5]. However, in practical situations, the lithium-ion battery capacity deteriorates in its life cycle due to various reasons such as loss of lithium-ion inventory, loss of electrolyte, loss of active materials and resistance increment [6–9]. An accurate capacity estimation will then help to estimate precisely the degradation status of those batteries, and finally benefit battery management system

(BMS) to ensure their safety and reliability [10,11].

In recent studies, a variety of data-driven methods have been developed for capacity estimation of lithium-ion batteries. Generally, these methods are mainly divided into three categories: differential analysis (DA) approaches [12–16], kernel-based approaches [17–21] and neural network approaches [22–26]. DA approaches are developed based on the simple mapping relationships between the characteristic values of differential charge/discharge capacity curves or voltage curves of the batteries and their fading capacities. Wang [13] investigated the differential voltage (DV) curves of LiFePO₄ batteries and found that the location intervals calculated between the first inflection and peak inflection of differential voltage curves exhibit a linear reduction trend with respect to the battery capacities. In addition, Zheng [15] studied the relationship between two inflection points on incremental capacity (IC) curves, one inflection point on DV curves and state of charge (SOC), and proposed an estimation method for the battery capacity using the cumulative charge obtained from the specific SOC intervals among these three inflection points. Although the DA approaches exhibit a high computational efficiency on the capacity estimation, their estimation accuracy in practical applications is questionable with a great deal of uncertainties. This is mainly because that IC and DV curves are sensitive with differential interval, and the chosen of feature points of IC or DV curves are also

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highly personalized.

In contrast to the DA approaches, kernel-based approaches provide a different way in capacity degradation estimation to avoid the differentiation of raw data. In this method, kernel-based learning algorithms such as support vector machine [18,20], relevance vector machine [17] and Gaussian process regression [19] are usually utilized to estimate the battery capacity by using characteristic values extracted from experimental results. Robert [19] proposed a Gaussian process regression method to estimate battery capacity based on small portions, rather than the full set, of voltage-time data under constant current charging and discharging conditions. His method offers special benefits to the circumstances with incomplete voltage-time data. Furthermore, application of optimization algorithms is capable to maintain a high level of robustness and practicability of the kernel-based learning algorithms. Zhang [17] demonstrated that by incorporating an accelerated particle swarm optimization algorithm into multi-kernel relevance vector machine method, the accuracy and robustness of the battery capacity estimation are obviously improved.

Recently, the battery capacity is estimated from a variety of characteristic features by using artificial neural network (ANN) techniques such as extreme learning machine [22], feedforward neural network [23], convolutional neural network (CNN) [27], long short-term memory network (LSTM) [26,28], Elman neural network (ELM) [29], etc. For instance, in Pan's study [22], ohmic internal resistances and polarized internal resistances of the batteries are extracted as health indicators for capacity estimation with an extreme learning machine technology. Dai [23] took 8 features including average voltage, voltage increment, etc. extracted from charging and discharging curves of the battery as prior knowledge and fed them to a feedforward neural network to estimate battery capacity. These above researches show that ANN

models can provide capacity estimations with good accuracy when the training data is sufficiently enough. Further, for coping with the situations with insufficient training data, Sheng [25] introduced the transfer learning and ensemble learning technologies into the battery capacity estimation to make full use of the experimental data collected from other batteries.

However, it has to be noted that almost all of these above-mentioned approaches are picky about the raw data. Some require large amount of data on voltage, current or time measurements in charge or discharge cycles to generate features such as the average voltage during constant-current charge or discharge process, etc., while others demand that the range of voltage, current or time measurement in cycles should cover specified intervals to achieve features including specified peaks in IC/DV curves, time values at equispaced voltage points within specified voltage range, voltage values at equispaced time points within specified voltage range, etc. However, in many practical situations, these two types of raw data are difficult to obtain due to the uncertainty of user behavior on charging and discharging schedules. According to a statistical survey of EV projects in Ref. [30], the batteries in vehicles are most likely to be charged start from its SOC laid between 20% and 80% and end up with SOC between 70% and 100%. To cope with this problem, it is necessary to develop a new battery capacity estimation method that is robustness for input data whenever acquired from practical usages.

In the field of CNN, one-dimensional convolutional neural networks (1D CNNs) are receiving more and more attention in condition monitoring, structural damage detection, etc., owing to their superiority on dealing with one-dimensional (1D) signals and low calculation cost [41]. Therefore, an 1D CNN based capacity estimation method is developed in this paper to estimate the lithium-ion battery capacity in real time from arbitrary segments of the

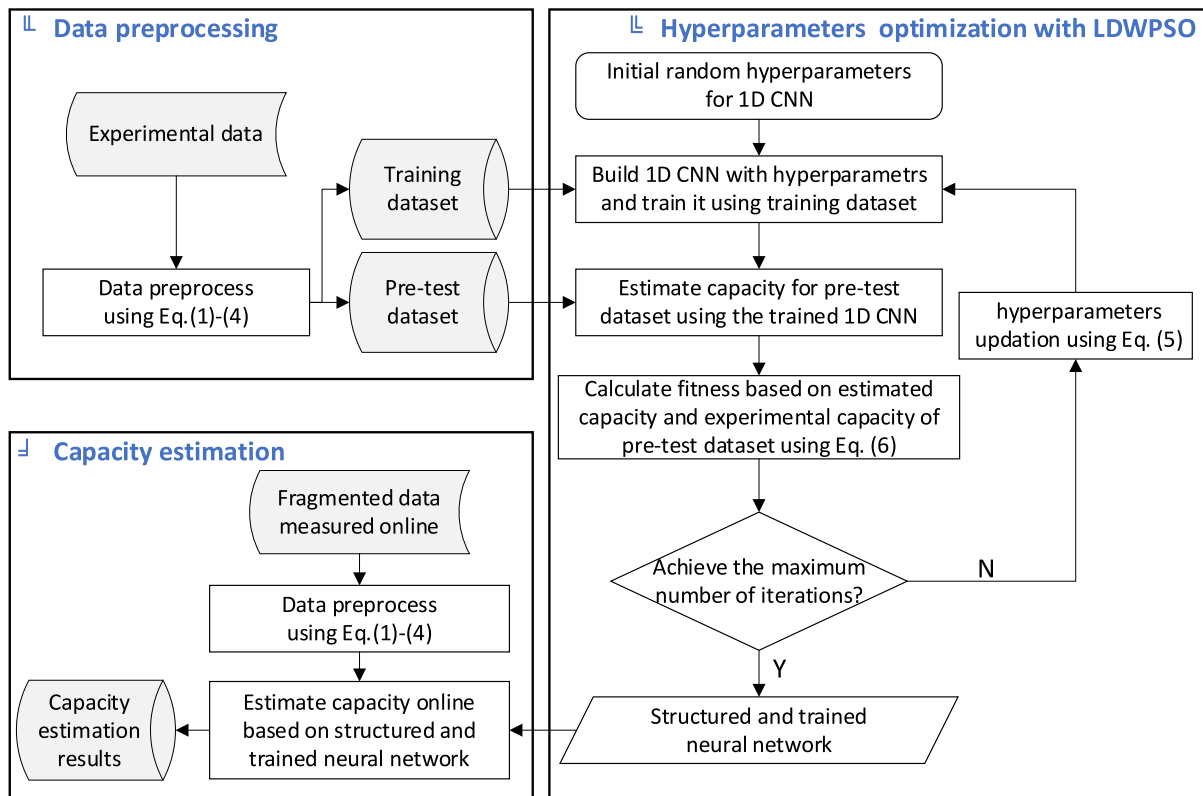


Fig. 1. Procedure of capacity estimation for lithium-ion batteries by using the proposed 1D CNN model.

battery charging curves. Moreover, since the performance of the 1D CNN model is highly affected by its hyperparameters, a linear decreasing weighted particle swarm optimization (LDWPSO) algorithm, in which velocities of particles are updated with linearly decreasing inertia factors to improve the global and local search abilities of particles [32,33], is ensemble to optimize the hyperparameters of the 1D CNN model. According to the research in Ref. [34], LDWPSO has been demonstrated with the capability to find excellent hyperparameters for neural networks.

The remainder of this paper is organized as follows. Section 2 illustrates the principle of the modeling methodology and presents the architecture of the proposed 1D CNN model. Section 3 introduces the experimental data for training and validating the proposed 1D CNN model. Section 4 provides a detailed discussion on capacity estimations and verifications. Finally, Section 5 draws the main conclusions.

2. Methodology

The procedure of capacity estimation based on the proposed 1D CNN model is illustrated in Fig. 1, including three steps which are data preprocessing, hyperparameters optimization and capacity estimation respectively. In the data preprocessing step, experimental data of voltages, differential voltages and currents measured from a couple of lithium-ion batteries under thousands of charge and discharge cycles are preprocessed into a training dataset and pre-test dataset respectively. In the hyperparameters optimization step, the key hyperparameters of the 1D CNN model are represented by particles in LDWPSO. The 1D CNN model is established based on those hyperparameters, and trained with the training dataset. Afterward, the pre-test dataset is used to evaluate the fitness of particles in LDWPSO and determine the search direction of particles in the next step. Such above iterative optimization loop stops when the iteration number reaches its maximum value, to finally obtain the optimized hyperparameters and the 1D CNN parameters in correspondence to these hyperparameters. In the capacity estimation step, with the optimized hyperparameters and trained parameters, the proposed 1D CNN model can be then used to estimate the capacity of the battery in real time based on the random segmented charge curves extracted from charge cycles.

2.1. 1D CNN model

Development of the 1D CNN model is the core task to achieve battery capacity estimation. As illustrated in Fig. 2(a), the 1D CNN model developed contains three types of operations, namely the 1D convolution, pooling and full connection. Among these operations, the 1D convolutions are utilized to detect features from input data; the pooling operation is used to reduce the data dimension for inputs of the fully-connected layers; and the full connection is supposed to learn dependences between high-level features obtained from the pooling layer and corresponding battery capacity. Specifically, the proposed 1D CNN model consists of three convolution layers, one global pooling layer and two fully-connected layers, as shown in Fig. 2(b). After each convolution layer, a batch normalization technique is applied to improve the performance and stability of the 1D CNN model [35]. Moreover, as listed in Table 1, the stride sizes in the convolutional layers and pooling layer are artificially set to 1, while other hyperparameters are supposed to be optimized with optimization algorithm. Before training, the 1D CNN parameters are initialized with a uniform distribution by using Xavier initialization method [36]. Then in the training process, these parameters are updated based on a gradient descent algorithm and mean square error loss function.

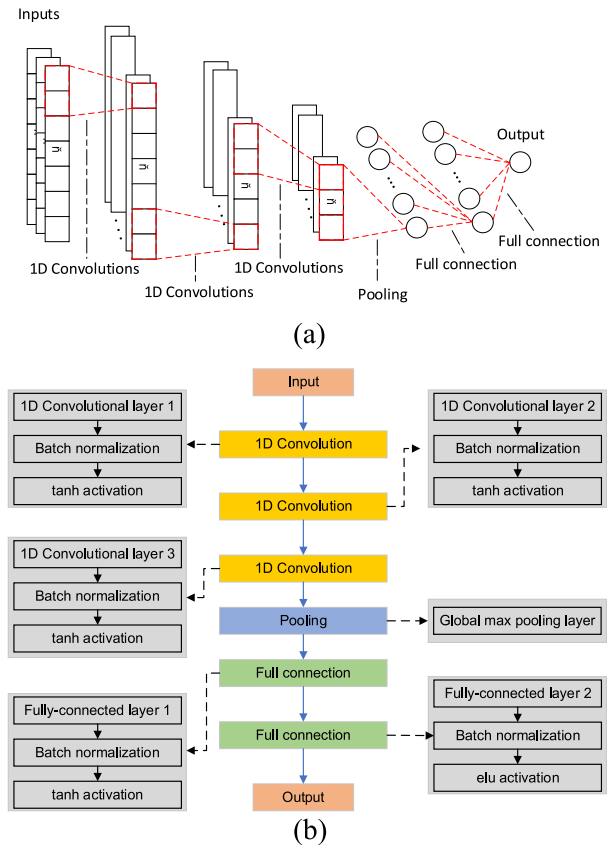


Fig. 2. Illustration of the proposed 1D CNN model: (a) model struct; (b) model architecture.

Table 1

Configurations of the proposed 1D CNN model.

Layer name	Filters/Units	Kernel size	Stride size
Convolutional layer 1 (Conv.1)	t.b.o ^a	t.b.o	1
Convolutional layer 2 (Conv.2)	t.b.o	t.b.o	1
Convolutional layer 3 (Conv.3)	t.b.o	t.b.o	1
Global max pooling	—	—	—
Fully-connected layer 1 (FC.1)	t.b.o	—	—
Fully-connected layer 2 (FC.2)	1	—	—

^a 't.b.o' indicates parameters to be optimized.

2.2. Input data for the 1D CNN model

In overall, the input data for the 1D CNN model is composed by three segmented parts simultaneously extracted from the charging voltage curve, differential charging voltage curve and charging current curve from a single charge cycle, as depicted by the red lines in Fig. 3. The differential voltage curve is approximately calculated as the gradient of charge voltage v with respect to charge capacity q by using Eq. (1).

$$\frac{dv}{dq} \approx \frac{\Delta v}{\Delta q} \quad (1)$$

Among the input data, the voltage is frequently used in many networks for capacity estimation of batteries [27,40]. And the differential voltage is one of the most important features in DA approaches for capacity estimation [13,15]. However, as can be seen from Fig. 3, in the period of constant voltage charging, the charging

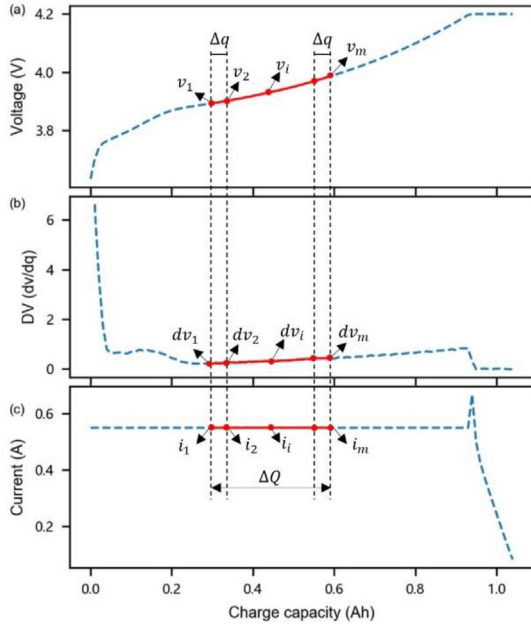


Fig. 3. Segmented data randomly selected from charge curves under a single charge cycle. (a): Segmented data from the charging voltage curve; (b): Segmented data from the differential charging voltage curve; (c): Segmented data from the charging current curve.

voltage and differential charging voltage become very insensitive with the charge capacity. This will significantly increase the difficulty to remain the modeling accuracy in a high level. For solving this problem, the charging current curve is involved into the input features as well.

By considering the charging uncertainties, segmented data in the three charging curves are randomly selected with the same charge capacity increment of ΔQ . Then each segmented data is discretized into m equally spaced samples in terms of a charge capacity increment of Δq using Eq. (2).

$$\begin{cases} \Delta Q = k \times C_{ini} \\ m = \frac{\Delta Q}{\Delta q} \end{cases} \quad (2)$$

where C_{ini} is the nominal capacity of the battery, k is the coefficient to represent the length of segments equaling to 0.35 in this paper unless specifically stated, $\lfloor x \rfloor$ is the floor function to export the greatest integer no more than x .

In addition, to improve the stability of the 1D CNN model and speed up the training process [38], all input data are normalized using Eq. (3) before utilization:

Table 2
Specifications of the test samples and test profiles.

Batteries	CS2-33	CS2-34	CS2-35	CS2-36	CX2-34	CX2-36	CX2-37	CX2-38
Dimensions (mm)	5.4*33.6*50.6				6.6*33.8*50			
Weight (g)	21.1	21.1	21.1	21.1	28	28	28	28
Rated capacity (Ah)	1.1	1.1	1.1	1.1	1.35	1.35	1.35	1.35
Constant charge current (A)	0.55	0.55	0.55	0.55	0.675	0.675	0.675	0.675
Maximum charge voltage (V)	4.2	4.2	4.2	4.2	4.2	4.2	4.2	4.2
End-of-charge current (A)	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Discharge cut-off voltage (V)	2.7	2.7	2.7	2.7	2.7	2.7	2.7	2.7
Discharge current (A)	0.55	0.55	1.1	1.1	1.35	1.35	1.35	1.35
Ambient temperature (°C)	25	25	25	25	25	25	25	25

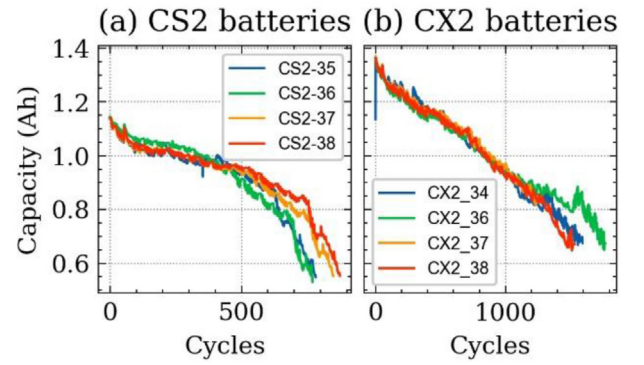


Fig. 4. Experimental capacity degradation curves of the test samples (a): CS2 batteries; (b): CX2 batteries.

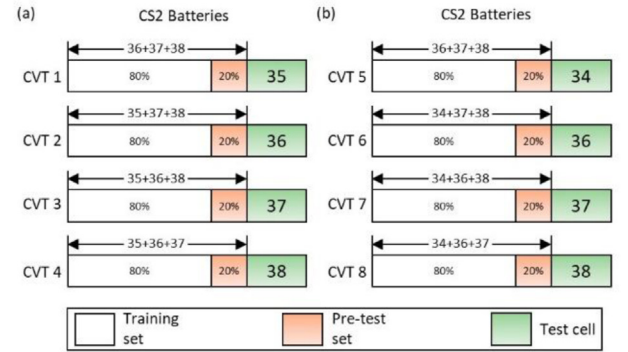


Fig. 5. Assignment strategy of the experimental data of batteries in the two groups for cross validation: (a) CS2 batteries; (b) CX2 batteries.

$$x_n = -1 + 2 \times \frac{x - \min(x_{train})}{\max(x_{train}) - \min(x_{train})} \quad (3)$$

where x_{train} is the voltage, differential voltage or current in the training dataset and x_n is the normalized result in terms of x .

At last, the segmented charge voltages, segmented differential charge voltages and segmented charge currents, are filled in a matrix X with a size of $m \times 3$, as illustrated in Eq. (4). Based on the input X , the 1D CNN, after well training, is capable to export the remaining capacity of the battery online.

$$X = [V \quad DV \quad I] = \begin{bmatrix} v_1 & dv_1 & i_1 \\ \vdots & \vdots & \vdots \\ v_i & dv_i & i_i \\ \vdots & \vdots & \vdots \\ v_m & dv_m & i_m \end{bmatrix}_{m \times 3} \quad (4)$$

2.3. Hyperparameters optimization

LDWPSO algorithm is employed to optimize the 't.b.o' hyperparameters of the 1D CNN listed in Table 1. The update equations of the position and velocity of particles used in LDWPSO are formulated as:

$$\begin{cases} x_i^{new} = x_i^{old} + v_i^{new} \\ v_i^{new} = w^k * v_i^{old} + c_1 * r_1 * (pbest_i^k - x_i^{old}) + c_2 * r_2 * (gbest^k - x_i^{old}) \\ w^k = w_{min} + (w_{max} - w_{min}) * \frac{Iter_{max} - k}{Iter_{max}} \end{cases} \quad (5)$$

where coordinate x_i of particle i represents the 't.b.o' hyperparameter, v_i is the velocity of particle i , r_1, r_2 are random values in between 0 and 1, c_1, c_2 are constant coefficients equaling to 2, $pbest_i^k, gbest^k$ are best encountered positions of particle i and all particles respectively at iteration k with smallest fitness, w_{min}, w_{max} are constant coefficients equaling to 0.4 and 0.9 respectively, and $Iter_{max}$ is the maximum iteration number. In this study, the population of particles, in LDWPSO were manually set to 20 and 200 respectively. And root mean squared error (RMSE) values calculated by Eq. (6) are taken as fitness to compare the particles in LDWPSO [27].

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (c_i^* - c_i)^2} \quad (6)$$

where c_i^*, c_i are estimated and experimental capacities in pre-test datasets respectively.

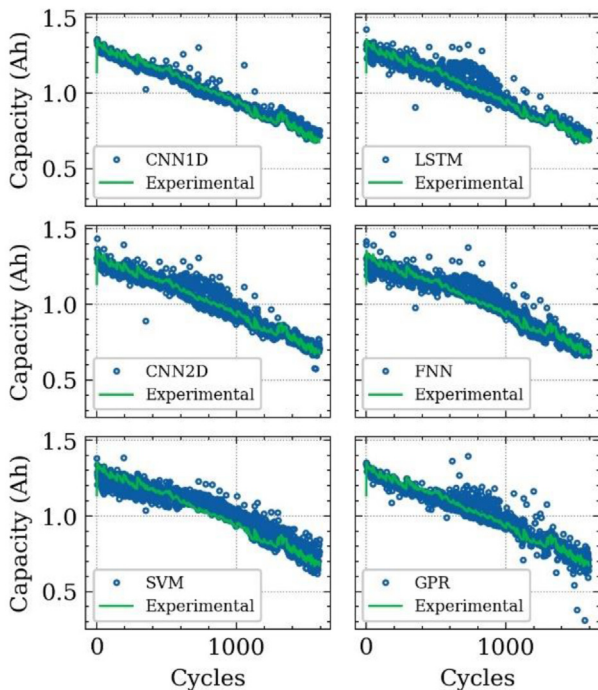


Fig. 6. Capacity estimation results of battery CX2-34 by different methods.

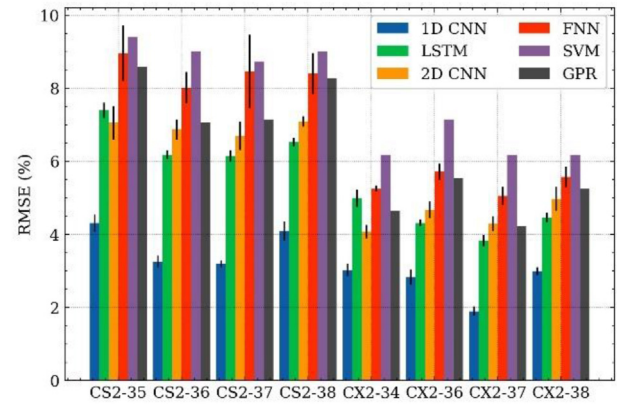


Fig. 7. RMSEs of capacity estimation results for 8 batteries using different methods.

Table 3

Values of m for CS2 and CX2 batteries with respect to different k .

Battery	k				
	0.3	0.35	0.4	0.45	0.5
CS2 batteries	33	38	44	49	55
CX2 batteries	40	47	54	60	67

3. Experimental data

Experimental results on two groups of different LiCoO₂ batteries tested by Center of Advanced Life Cycle Engineering (CALCE) of University of Maryland are obtained from Ref. [39], and employed in this study. 4 CS2 lithium-ion batteries of 1.1Ah rated capacity (named by CS2-35, CS2-36, CS2-37, CS2-38 respectively) and 4 CX2 lithium-ion batteries of 1.35Ah rated capacity (named by CX2-34, CS2-36, CS2-37, CS2-38 respectively) were tested under the same testing conditions with an Arbin Battery Tester. All batteries are cycled with constant current-constant voltage (CC-CV) charging mode and discharged at a fixed current. During each charging process, all batteries are firstly charged with a constant current rate of 0.5C until the voltage reached 4.2V. Then the charged voltage is remained constantly as 4.2V until the charging current dropped to below 0.05 A. And the cut off voltage in discharge stage is set as 2.7V. Detailed specifications about the test samples and test profiles are listed in Table 2. The measured battery capacity is calculated by integrating the discharge current over time for the entire discharge cycle, as suggested in the coulomb counting method. Fig. 4 shows the experimental capacity degradation curves of all batteries in the two groups.

The estimation accuracy of the proposed 1D CNN model is cross validated based on the experimental data from the CS2 and CX2

Table 4

Optimized hyperparameters for the proposed 1D CNN model with respect to different k .

Items	k				
	0.3	0.35	0.4	0.45	0.5
Conv.1 filters	17	44	25	60	11
Conv.1 kernel size	6	7	2	9	10
Conv.2 filters	14	42	3	60	60
Conv.2 kernel size	3	4	2	1	1
Conv.3 filters	27	54	30	60	56
Conv.3 kernel size	19	16	23	17	14
FC.1 units	39	60	42	55	60

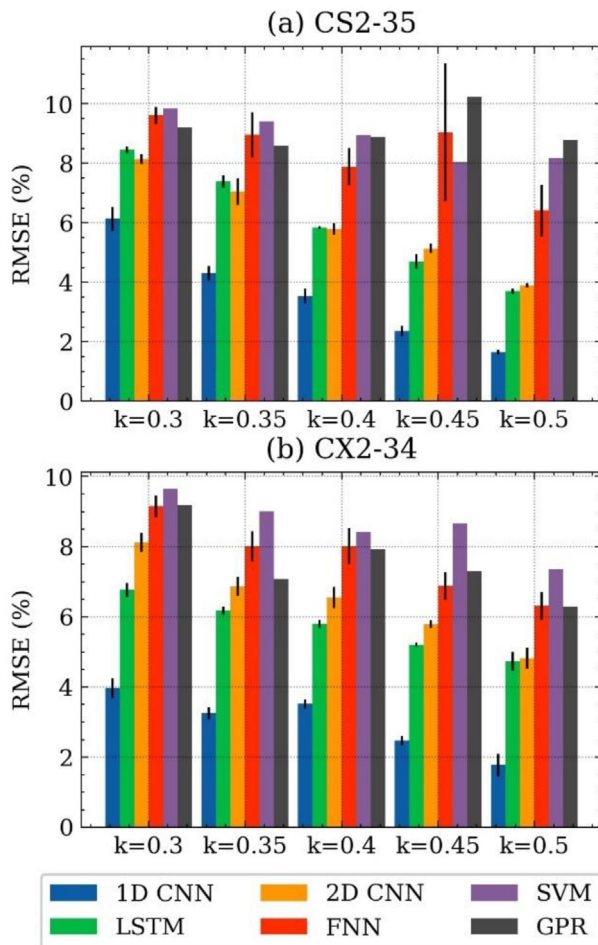


Fig. 8. RMSEs of capacity estimation results from the batteries CS2-35 and CX2-34 with respect to different k .

battery groups, as illustrated in Fig. 5. In each group, each of the 4 batteries was picked as the test battery, whereas the rest three are regarded as the source batteries for generating source datasets. Therefore, in total there are 8 cross-validation trials (CVTs) created by the experimental data from the two battery groups, as shown in Fig. 5. In addition, in order to make full use of the source batteries to obtain a large source dataset, for all source batteries in each CVT, charge voltage curve, differential voltage curve and charge current curve at every cycle are sliced into multiple data segments of length ΔQ , of which the initial charge capacities are shifted by 0.05 A one after another. As a result, the number of samples in the source dataset of each CVT is greater than 20000. Then, all samples in source dataset are randomly shuffled, and divided into training dataset and pre-test dataset by the ratio of 8:2. For the test battery, the segmented charge curves randomly selected from each charge cycle are imported to the 1D CNN model to estimate the battery capacity, for the purpose of the 1D CNN model validation.

4. Results and discussions

In this study, capacity estimations by using the proposed 1D CNN model were carried out with a desktop PC equipped with an Intel i9 7980XE CPU and a RTX2080 TI-A11G graphics processing unit (GPU). The epochs and batch size are set to 100 and 256 respectively for training the 1D CNN. It approximately takes 50–100 s to complete the training process in every CVT. In addition,

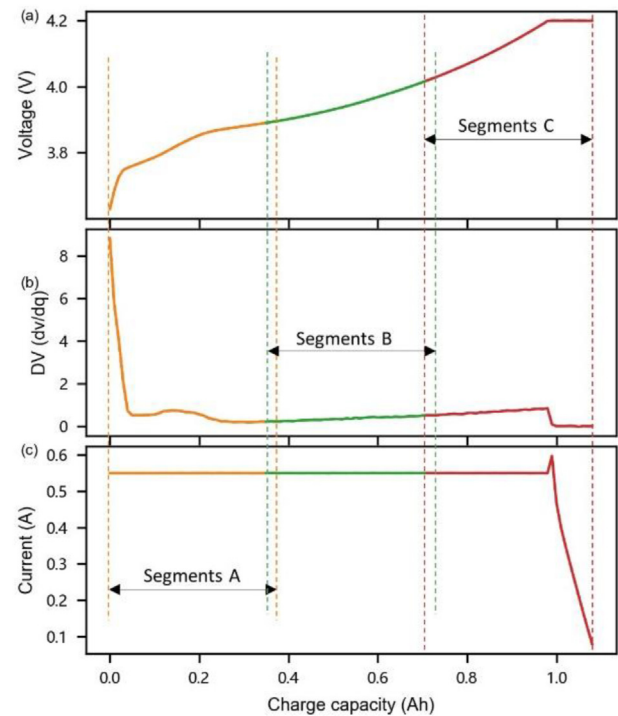


Fig. 9. Segments in different positions.

from the comparative perspective, capacity estimations with randomly segmented data as inputs are also addressed using other existing neural network models including long short-term memory neural network (LSTM), two-dimensional convolutional neural network (2D CNN) and feed-forward neural network (FNN), and kernel-based methods including support vector machine (SVM) and Gaussian process regression (GPR). And four neural networks and two kernel-based methods were implemented in python using Keras module and scikit-learn module respectively.

4.1. Capacity estimation results

In order to minimize effects of random initialization of network parameters on the capacity estimation accuracy, 30 different capacity estimation results of each neural network for every test battery are obtained by using the Monte Carlo method. Shown as an example, the capacity estimation results of the battery CX2-34 in one of the 30 runs are plotted in Fig. 6. It can be seen that the 1D CNN provides the best estimations on the battery capacities in the entire degradation process, compared to the LSTM, 2D CNN, FNN, SVM and GPR methods. This phenomenon is proved by Fig. 7 where RMSEs of capacity estimation results by the 1D CNN model are much lower than those by the other 5 methods. The poorest capacity estimations are from the SVM, whereas the capacity estimations from the other 4 methods for comparison are slightly more or less better, but still far worse than the 1D CNN model. Those results suggest that the proposed 1D CNN method has the unique advantage on capacity estimations based on random segments of charging curves.

4.2. Effects of segment lengths on capacity estimation

To further investigate the effect of length of random segments on the capacity estimation results, different source datasets consisting of segmented curves with a varied k value from 0.3 to 0.5

Table 5
Positions of 3 different segmented curves.

Segmented curves ID	Initial SOC	End of SOC
Segments A	0%	kC_i/C_{ini}
Segments B	$50\% - kC_i/2C_{ini}$	$50\% + kC_i/2C_{ini}$
Segments C	$100\% - kC_i/C_{ini}$	100%

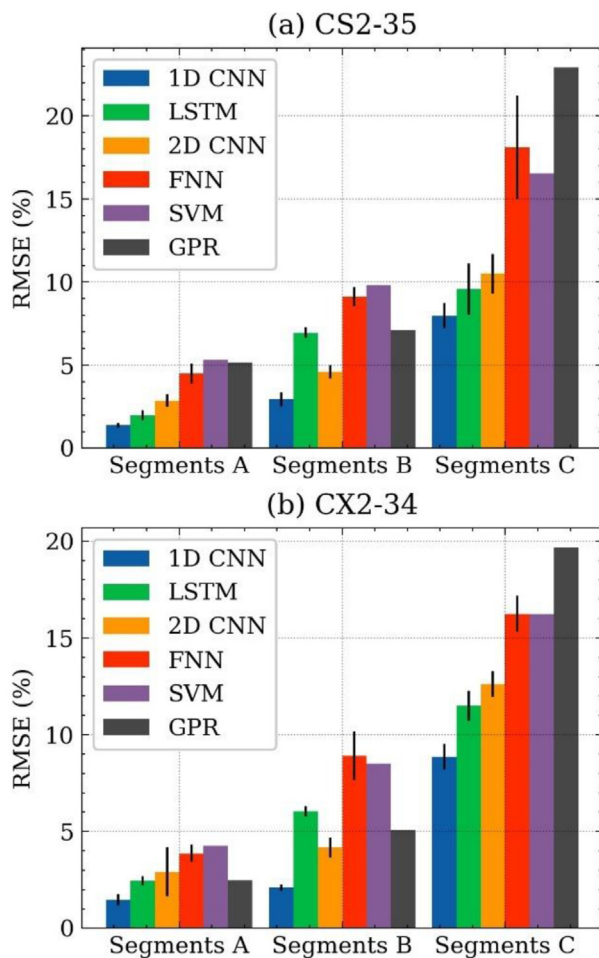


Fig. 10. RMSEs of capacity estimation results of batteries CS2-35, CX2-34 with segments at different positions.

were generated for each CVT. Furthermore, the values of m for CS2 and CX2 batteries with respect to different k are provided in Table 3. And the hyperparameters of 1D CNN optimized by LDWPSO at different k are listed in Table 4. For different datasets, each network is trained 30 times independently for each dataset as well.

Fig. 8 shows the RMSEs of capacity estimations from the batteries CS2-35, CX2-34 by the 6 methods with respect to the different segment length k . It can be seen that the capacity estimation RMSEs by the proposed 1D CNN are the smallest among the 6 methods, in all of the situations with different k values. In addition, for any of the 6 methods, the RMSEs of capacity estimations show a decreasing trend with the increase of k in general. This indicates that the longer segment contributes to a high accuracy of capacity estimation. For the other 6 batteries, similar trends on RMSEs of the capacity estimations with respect to either different methods or different segment lengths are also observed to prove that the above two observations are not accidental phenomena.

4.3. Effects of segment positions on capacity estimation

Because of the high nonlinear characteristic of the charging curve, the estimated capacity shows dependency on the input segmentation data extracted from different locations of the charging curve. In this section, segments of the charging voltage, differential charging voltage and charging current curves are sliced from 3 different positions (i.e. segments A, segments B and segments C as shown in Fig. 9) as input data for networks to investigate the effect of segment positions on battery capacity estimations. Specifically, the initial and end of SOCs of segments A, segments B and segments C are listed in Table 5, where C_i is the battery capacity at current state.

RMSEs of the estimation results of batteries CS2-35 and CX2-34 using segmentation data from Segments A, Segments B and Segments C as inputs are plotted and compared in Fig. 10. It can be seen that all of the 6 methods give increased RMSEs with the movement of the segment to the end of the charging curves. Especially when the segmentation data are selected from the last part of charging curves, RMSEs of all of the network models are larger than 7.5%. This implies that the segmented curves should be extracted as early as possible in charge process to improve the capacity estimation accuracy in practical applications. Meanwhile, as can be seen from Fig. 10, among the 6 methods, the proposed 1D CNN model still provides the best capacity estimations for all scenarios of input segmented curves.

5. Conclusions

In this study, a 1D CNN model is proposed for capacity estimation of lithium-ion batteries. Considering the uncertainty of the charging behavior in actual usage, random segments of the charging voltage curve, differential charging voltage curve and charging current curve are used as the input data of the 1D CNN model. And LDWPSO is utilized in key hyperparameters optimization of the 1D CNN model. According to the estimated results, the proposed 1D CNN model demonstrates more superior performance than a number of existing methods such as the LSTM, 2D CNN, FNN, SVM and GPR when using the above-mentioned arbitrary segments of charging curves as inputs. A further study shows that the capacity estimation accuracy can be improved by either extending the length of segment curves or picking up the early parts of the charging curves as inputs of the 1D CNN model.

Credit author statement

Cheng Qian: Writing – review & editing, Formal analysis, Binghui Xu: Conceptualization, Methodology, Investigation, Software, Writing – original draft, Liang Chang: Investigation, Formal analysis, Bo Sun: Validation, Visualization, Qiang Feng: Resources, Software, Dezhen Yang: Data curation, Project administration, Yi Ren: Funding acquisition, Project administration, Zili Wang: Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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